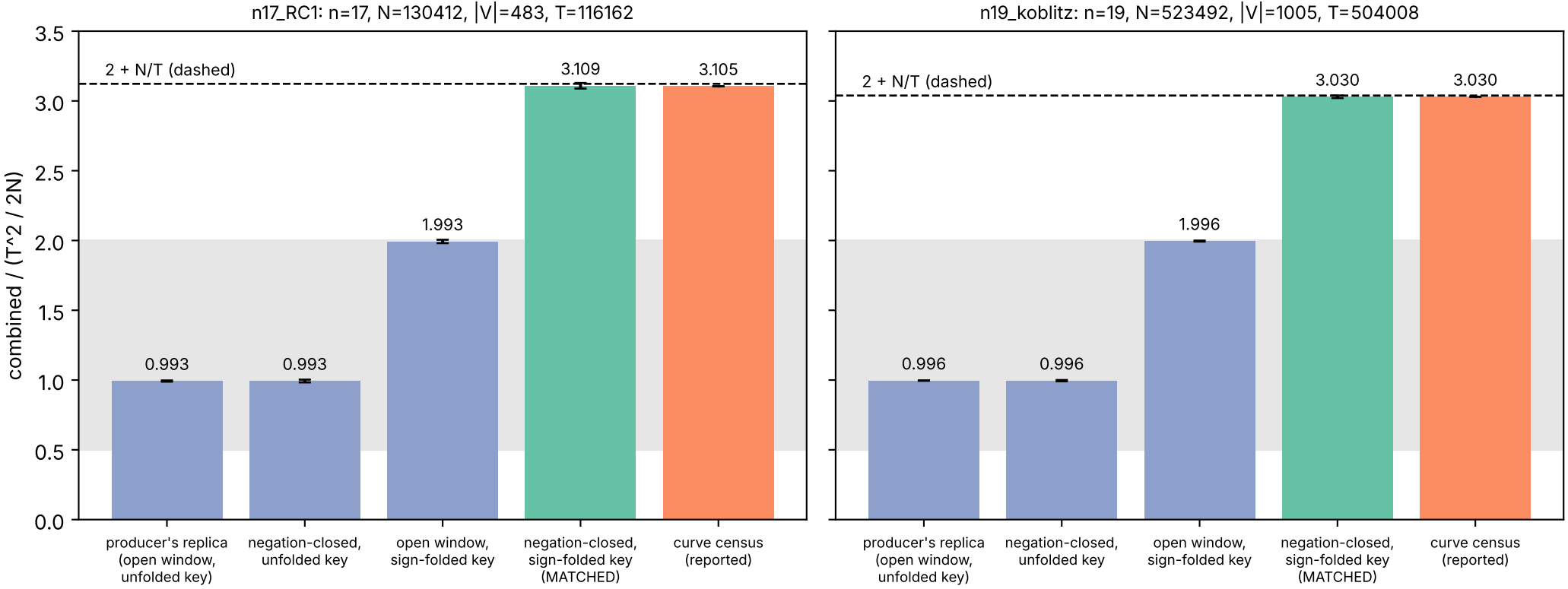
EV-BINSTD-769c7c: collision ratio under matched generic nulls (bars: mean of 20 windows +/- sd; grey band: the DO-1 band [0.5, 2])
n17_RC1: n=17, N=130412, |V|=483, T=116162
n19_koblitz: n=19, N=523492, |V|=1005, T=504008
combined / (T^2 / 2N)
2 + N/T (dashed)
0.993
0.993
1.993
3.109
3.105
0.996
0.996
1.996
3.030
3.030
producer's replica (open window, unfolded key)
negation-closed, unfolded key
open window, sign-folded key
negation-closed, sign-folded key (MATCHED)
curve census (reported)

```
== n17_RC1: GF(2^17), modulus 0x20009, a=97044, b=126251, #E=130412, prefix window deg(x)<9
   checks: affine points + 1 = 130412 (= #E: True); N*P = O on 6 random points: True; vectorised x(P+Q) = scalar law on 200 pairs: True
   reproduction of the producer's census: |V|=483 T=116162 identity-sum pairs=241 combined=160633 distinct keys=38445 max multiplicity=16 reported-style ratio=3.104947 S_fit=42001.36
   trivial partner collisions {P,Q} vs {-P,-Q}: 58081 (= T/2 = 58081; self-partner pairs 0)
   nontrivial collisions 102552 vs expected 103467.7 (uniform sign-folded keys): ratio 0.9911; predicted reported-style ratio 3.1226 (2+N/T = 3.1227)
   generic, open window, unfolded key (producer's replica design)     ratio 0.9928 +/- 0.0043 (min 0.9872, max 1.0029, 20 windows)
   generic, negation-closed window, unfolded key                       ratio 0.9931 +/- 0.0100 (min 0.9695, max 1.0131, 20 windows)
   generic, open window, sign-folded key                               ratio 1.9931 +/- 0.0122 (min 1.9638, max 2.0204, 20 windows)
   generic, negation-closed window, sign-folded key (MATCHED)          ratio 3.1088 +/- 0.0199 (min 3.0616, max 3.1488, 20 windows)
   curve nontrivial 102552 vs matched generic 102749.8 +/- 1031.5: z = -0.19, ratio 0.9981
   curve window translated_prefix      |V|=488   T=118584  reported-style ratio 3.0625 (predicted 3.0997); nontrivial/expected 0.9814
   curve window random_subspace_dim_k  |V|=523   T=136242  reported-style ratio 2.9378 (predicted 2.9572); nontrivial/expected 0.9903
   curve window random_xset_size_2^k   |V|=516   T=132612  reported-style ratio 2.9712 (predicted 2.9834); nontrivial/expected 0.9939
   8 random x-set windows of 2^k abscissae: nontrivial/expected 0.9935 +/- 0.0106 (min 0.9810, max 1.0096); reported-style ratio 2.999 (min 2.876, max 3.212)
   curve, one point per abscissa (no negation closure), |V|=503: reported-style ratio 1.9922; combined/expected (sign-folded) 0.9961
   aligned half V'_0 (constant term of x = 0): |V|=245 T=29768 combined=28436 reported-style ratio 8.3698 (producer 8.37); points in 2G: 245/245, complement half in 2G: 0/238; nontrivial/expected if confined to 2G 0.9973; predicted 8.3807; generic confined window 8.306 +/- 0.100

== n19_koblitz: GF(2^19), modulus 0x80027, a=0, b=1, #E=523492, prefix window deg(x)<10
   checks: affine points + 1 = 523492 (= #E: True); N*P = O on 6 random points: True; vectorised x(P+Q) = scalar law on 200 pairs: True
   reproduction of the producer's census: |V|=1005 T=504008 identity-sum pairs=502 combined=735192 distinct keys=161827 max multiplicity=18 reported-style ratio=3.030163 S_fit=172760.36
   trivial partner collisions {P,Q} vs {-P,-Q}: 252004 (= T/2 = 252004; self-partner pairs 0)
   nontrivial collisions 483188 vs expected 485247.7 (uniform sign-folded keys): ratio 0.9958; predicted reported-style ratio 3.0387 (2+N/T = 3.0387)
   generic, open window, unfolded key (producer's replica design)     ratio 0.9960 +/- 0.0014 (min 0.9932, max 0.9984, 20 windows)
   generic, negation-closed window, unfolded key                       ratio 0.9955 +/- 0.0046 (min 0.9856, max 1.0019, 20 windows)
   generic, open window, sign-folded key                               ratio 1.9958 +/- 0.0036 (min 1.9881, max 2.0022, 20 windows)
   generic, negation-closed window, sign-folded key (MATCHED)          ratio 3.0297 +/- 0.0092 (min 3.0099, max 3.0424, 20 windows)
   curve nontrivial 483188 vs matched generic 483073.0 +/- 2240.2: z = +0.05, ratio 1.0002
   curve window translated_prefix      |V|=1048  T=548104  reported-style ratio 2.9617 (predicted 2.9551); nontrivial/expected 1.0033
   curve window random_subspace_dim_k  |V|=961   T=460800  reported-style ratio 3.1248 (predicted 3.1360); nontrivial/expected 0.9944
   curve window random_xset_size_2^k   |V|=984   T=483144  reported-style ratio 3.0840 (predicted 3.0835); nontrivial/expected 1.0003
   8 random x-set windows of 2^k abscissae: nontrivial/expected 0.9967 +/- 0.0051 (min 0.9910, max 1.0060); reported-style ratio 3.001 (min 2.871, max 3.095)
   curve, one point per abscissa (no negation closure), |V|=1026: reported-style ratio 1.9918; combined/expected (sign-folded) 0.9959

wrote output.json
```